Sungjin Cheong

Ph.D. Student in Computer Sciences, University of Wisconsin-Madison

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Education

- **University of Wisconsin-Madison** — *Sep. 2025 – Present*
  Ph.D. in Computer Sciences (*Advisor: Mohit Gupta*) — Madison, WI, USA

- **KAIST** — *Sep. 2021 – Aug. 2023*
  M.S. in Mechanical Engineering (*Advisor: Yong-Hwa Park*) — Daejeon, Republic of Korea

- **Yonsei University** — *Mar. 2014 – Aug. 2021*
  B.S. in Mechanical Engineering — Seoul, Republic of Korea

Research Interests

- Multisensor World Modeling and 3D Scene Representation
- Computational Imaging and ToF/LiDAR Sensing
- Inverse Problems and Statistical Signal Processing
- Physics-Based and Learning-Based Modeling

Publications & Conferences

J = Journal, IC = International Conference, DC = Domestic Conference

[IC.1] **Ghosts in the Point Clouds: De-glaring LiDAR in the Transient Domain**
Gump, A.; Henley, C. A.; **Cheong, S. J.**; Prabhakara, A.; Gupta, M. *CVPR*

[J.1] **Thermal-Infrared Remote-Target Detection System for Maritime Rescue Using 3-D Game-Based Data Augmentation With GAN**
Cheong, S. J.; Jung, W.; Lim, Y. S.; Park, Y. H.
IEEE Transactions on Geoscience and Remote Sensing (TGRS) (SCI, IF: 8.6)

[IC.2] **LiDAR Blooming Artifacts Estimation Method Induced by Retro-Reflectance With Synthetic Data Modeling and Deep Learning**
Cheong, S. J.; Ha, J. S. *ICCE-Asia (Oral, Accepted)*

[IC.3] **Infrared Image-Based Remote Target Detection for Maritime Rescue Utilizing a Deep Learning Network and Data Augmentation**
Cheong, S. J.; Lim, Y. S.; Jung, W. H.; Park, Y. H. *SPIE (Oral)*

[IC.4] **MEMS-Based Indirect Time-of-Flight Scanning LiDAR With Parallel-Phase Demodulation and Multi-path Interference Suppression**
Lee, S. H.; Lim, Y. S.; **Cheong, S. J.**; Park, Y. H. *SPIE*

[IC.5] **A Long-Distance 3D Face Recognition Architecture Utilizing MEMS-Based Region-Scanning LiDAR**
Lim, Y. S.; Lee, S. H.; **Cheong, S. J.**; Park, Y. H. *SPIE*

[DC.1] **Head Pose Estimation Method Using Depth Camera Based on 3D Point Cloud**
Cheong, S. J.; Lee, S. H.; Lim, Y. S.; Park, Y. H. *The Korean Society of Mechanical Engineers (Oral)*

[DC.2] **Validation of Simulation Model for 3D Indirect Time-of-Flight Sensor**
Lee, S. H.; Lim, Y. S.; **Cheong, S. J.**; Park, Y. H. *The Korean Society of Mechanical Engineers*

[DC.3] **Infrared Small Object Detection Based on Histogram Transformation**
Cheong, S. J.; Jung, W.; Park, Y. H. *The Korean Society of Mechanical Engineers*

[DC.4] **Depth Image Super Resolution Method for Time-of-Flight Camera Using Single RGB Image Depth Estimation**
Lim, Y. S.; Lee, S. H.; **Cheong, S. J.**; Park, Y. H. *The Korean Society of Mechanical Engineers*

[DC.5] **Depth Error Correction of 3D Scanning Sensor Using Machine Learning**
Lee, S. H.; Lim, Y. S.; **Cheong, S. J.**; Park, Y. H. *The Korean Society of Mechanical Engineers*

Research & Work Experience

- **LG Innotek Inc.** *Aug. 2023 – Aug. 2025*
  LiDAR Software Researcher | Base Technology Lab Seoul, Republic of Korea

  Topic: Single-Photon LiDAR Imaging & Statistical Signal Processing

  ◦ Developed **statistical image and signal processing algorithms** for single-photon LiDAR.
  ◦ Designed pipelines to **suppress single-photon LiDAR artifacts**.
  ◦ Implemented **range-walk compensation** using statistical analysis of time-resolved photon histograms.

- **Korea Advanced Institute of Science and Technology (KAIST)** *Sep. 2021 – Aug. 2023*
  Graduate Research Assistant | Human-Machine Interaction Lab Daejeon, Republic of Korea

  Project: Indirect Time-of-Flight LiDAR Imaging & Signal Processing *(Intek Plus Inc.)*

  ◦ Contributed to optical design, modulation, and signal processing for an **iToF AMCW LiDAR platform**.
  ◦ Developed **ML-based signal processing algorithms** to compensate for **internal stray light and multipath interference** in iToF measurements.
  ◦ Built an **iToF LiDAR vision platform** combining depth and intensity data for 3D perception and human-machine interaction.

  Project: Thermal-Infrared Imaging *(Korean Coast Guard)*

  ◦ Led the development of an **infrared imaging system** for remote target detection, using **physics-based simulation** to augment training data and improve detection and localization.

Honors & Awards

- **Summer Research Fellowship**, University of Wisconsin-Madison *2026*
- **Academic Excellence Scholarship**, Yonsei University *2017, 2020*
- **Academic Excellence Award**, Yonsei University *2017*

Skills

- **Programming Languages:** Python, C/C++, MATLAB, LabVIEW
- **Frameworks, Libraries & Tools:** PyTorch, OpenCV, Open3D, MMDetection3D, ROS, Git
- **Visualization Tools:** CloudCompare, MeshLab, Weights & Biases
- **Hardware:** Oscilloscopes, function generators
- **Languages:** Korean (native), English (proficient); *IELTS Academic: 7.5/9.0 (L: 8.5, R: 8.0, W: 6.5, S: 7.0)*

Relevant Coursework

- **ML & Deep Learning:** Mathematical Foundation of ML, Advanced Deep Learning in NLP, Pattern Recognition, Theoretical Foundation of ML, Matrix Methods in ML
- **Computer Vision:** Mobile Robotics and Autonomous Navigation, Autonomous Mobile System Programming, Introduction to Visual Intelligence
- **Robotics:** Intelligent Robotics
- **Statistics & Signal Processing:** Probability and Statistics, Random Data Analysis and Processing
- **Optimization:** Nonlinear Optimization

Teaching Experience

- **University of Wisconsin-Madison** *Sep. 2025 – Present*
  Teaching Assistant Madison, WI, USA

  ◦ CS 766: Computer Vision
  ◦ CS 566: Introduction to Computer Vision